

Nonlinear Assimilation via Score-based Sequential Langevin Sampling

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Applications of Data Assimilation

Numerical weather prediction (NWP) uses **physical models** of the atmosphere and oceans to predict the weather based on the **measurement data**.

Physical models

Governing equations of atmosphere:

- ▶ conservation of momentum
- ▶ conservation of mass
- ▶ conservation of energy
- ▶ the equation of state for ideal gases
- ▶ conservation equation for water mass
- ▶ ...

Measurement data



Problem Formulation

Dynamics model: A discrete time **unobserved** stochastic process $(\mathbf{X}_k)_{k \geq 1}$ satisfying

$$\mathbf{X}_{k+1} = \mathcal{F}_k(\mathbf{X}_k, \mathbf{V}_k)$$

$$\rho_k(\mathbf{X}_{k+1}|\mathbf{X}_k) = p(\mathbf{X}_{k+1}|\mathbf{X}_k)$$

- ▶ $\mathcal{F}_k$ is a **known** time-dependent model.
- ▶ $\mathbf{V}_k$ is a random variable with **known** distribution.

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Measurement model: A discrete time **observed** stochastic process $(\mathbf{Y}_k)_{k \geq 1}$ satisfying

$$\mathbf{Y}_k = \mathcal{G}_k(\mathbf{X}_k, \mathbf{W}_k)$$

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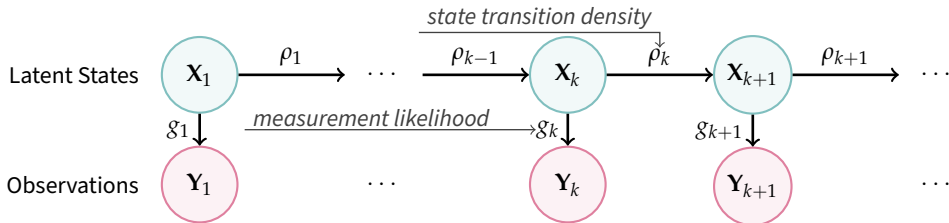
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Data Assimilation

The goal of the data assimilation is to estimate the posterior distribution of the latent state $\mathbf{X}_{k+1}$ given all available measurements $\mathbf{Y}_{[k+1]}$, that is, $\pi_{k+1}(\mathbf{X}_{k+1}|\mathbf{Y}_{[k+1]})$ for each $k \geq 1$.

Recursive Bayesian Filtering Framework

State-Space Model

Dynamics model	$\mathbf{X}_{k+1} = \mathcal{F}_k(\mathbf{X}_k, \mathbf{V}_k)$	$\longleftrightarrow$	State transition density	$\rho_k(\mathbf{X}_{k+1} \mathbf{X}_k)$
Measurement model	$\mathbf{Y}_{k+1} = \mathcal{G}_{k+1}(\mathbf{X}_{k+1}, \mathbf{W}_k)$	$\longleftrightarrow$	Measurement likelihood	$g_{k+1}(\mathbf{Y}_{k+1} \mathbf{X}_{k+1})$

$$\pi_{k+1}(\mathbf{X}_{k+1}|\mathbf{Y}_{[k+1]})$$

$$\propto p(\mathbf{Y}_{k+1}|\mathbf{X}_{k+1}, \mathbf{Y}_{[k]})q_{k+1}(\mathbf{X}_{k+1}|\mathbf{Y}_{[k]})$$

$$= p(\mathbf{Y}_{k+1}|\mathbf{X}_{k+1}, \mathbf{Y}_{[k]}) \int p(\mathbf{X}_{k+1}|\mathbf{X}_k, \mathbf{Y}_{[k]})\pi_k(\mathbf{X}_k|\mathbf{Y}_{[k]})d\mathbf{X}_k$$

$$= g_{k+1}(\mathbf{Y}_{k+1}|\mathbf{X}_{k+1}) \int \rho_k(\mathbf{X}_{k+1}|\mathbf{X}_k)\pi_k(\mathbf{X}_k|\mathbf{Y}_{[k]})d\mathbf{X}_k$$

► Bayes' rule

► Chapman-Kolmogorov identity

► $\mathbf{Y}_{k+1} \perp\!\!\!\perp \mathbf{Y}_{[k]}|\mathbf{X}_{k+1}$, and $\mathbf{X}_{k+1} \perp\!\!\!\perp \mathbf{Y}_{[k]}|\mathbf{X}_k$

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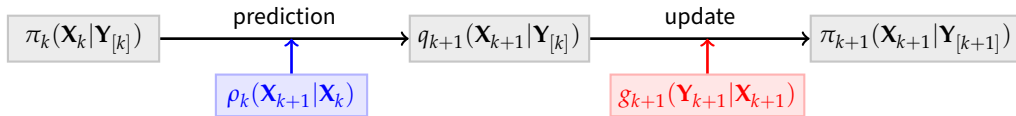
$$\overbrace{\pi_{k+1}(\mathbf{X}_{k+1}|\mathbf{Y}_{[k+1]})}^{\text{current posterior}} \propto \underbrace{g_{k+1}(\mathbf{Y}_{k+1}|\mathbf{X}_{k+1})}_{\text{measurement likelihood}} \underbrace{\int \underbrace{\rho_k(\mathbf{X}_{k+1}|\mathbf{X}_k)}_{\text{state transition}} \underbrace{\pi_k(\mathbf{X}_k|\mathbf{Y}_{[k]})}_{\text{previous posterior}} d\mathbf{X}_k}_{\text{current prior}}$$

Recursive Bayesian Filtering Framework

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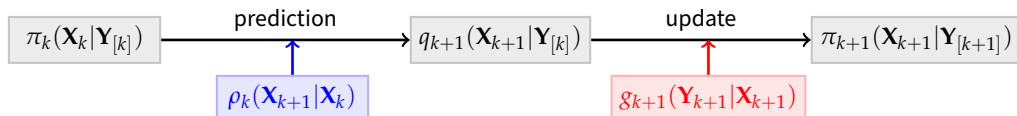
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Classical Methods for Data Assimilation

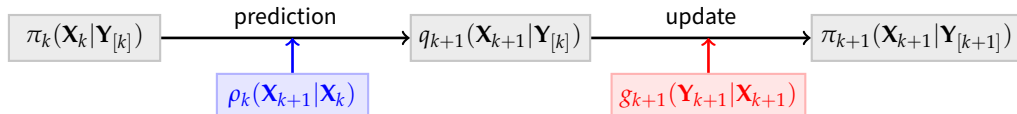
$$\underbrace{\pi_{k+1}(\mathbf{X}_{k+1}|\mathbf{Y}_{[k+1]})}_{\text{current posterior}} \propto \underbrace{g_{k+1}(\mathbf{Y}_{k+1}|\mathbf{X}_{k+1})}_{\text{likelihood}} \int \underbrace{\rho_k(\mathbf{X}_{k+1}|\mathbf{X}_k)}_{\text{transition density}} \underbrace{\pi_k(\mathbf{X}_k|\mathbf{Y}_{[k]})}_{\text{previous posterior}} d\mathbf{X}_k$$



- How to estimate the prediction distribution $q_{k+1}(\mathbf{X}_{k+1}|\mathbf{Y}_{[k]})$?
- How to sample from the posterior distribution $\pi_{k+1}(\mathbf{X}_{k+1}|\mathbf{Y}_{[k+1]})$ given the prediction and likelihood?

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- How to estimate the prediction distribution $q_{k+1}(\mathbf{X}_{k+1} | \mathbf{Y}_{[k]})$?
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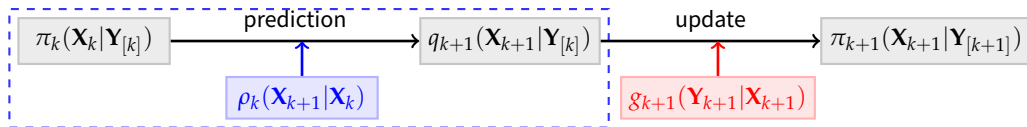
	$\pi_k(\mathbf{X}_k \mathbf{Y}_{[k]})$	$\rho_k(\mathbf{X}_{k+1} \mathbf{X}_k)$	$g_{k+1}(\mathbf{Y}_{k+1} \mathbf{X}_{k+1})$	Limitations
Kalman Filter	Gaussian approximation	Gaussian	Gaussian	linear + Gaussian
Particle Filter	particle approximation	–	–	particle degeneracy

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Prediction and Score Matching

Prediction density

$$q_{k+1}(\mathbf{X}_{k+1}|\mathbf{Y}_{[k]}) = \int \rho_k(\mathbf{X}_{k+1}|\mathbf{X}_k) \pi_k(\mathbf{X}_k|\mathbf{Y}_{[k]}) d\mathbf{X}_k$$



How to estimate the score function $\nabla \log q_{k+1}(\cdot|\mathbf{Y}_{[k]})$ of the prediction distribution?

$$q_{k+1}(\mathbf{X}_{k+1}|\mathbf{Y}_{[k]}) = \int \rho_k(\mathbf{X}_{k+1}|\mathbf{X}_k)\pi_k(\mathbf{X}_k|\mathbf{Y}_{[k]})d\mathbf{X}_k$$

- ▶ Given: $\hat{\mathbf{X}}_k^1, \dots, \hat{\mathbf{X}}_k^n \sim \text{i.i.d.}$ $\hat{\pi}_k(\mathbf{X}_k|\mathbf{Y}_{[k]}) \approx \pi_k(\mathbf{X}_k|\mathbf{Y}_{[k]})$.
- ▶ Prediction with **dynamics model**: $\underline{\mathbf{X}}_{k+1}^i = \mathcal{F}_k(\hat{\mathbf{X}}_k^i, \mathbf{V}_k^i)$, for each $1 \leq i \leq n$.

$$\underline{\mathbf{X}}_{k+1}^i \sim \hat{q}_{k+1}(\mathbf{X}_{k+1}|\mathbf{Y}_{[k]}) := \int \rho_k(\mathbf{X}_{k+1}|\mathbf{X}_k)\hat{\pi}_k(\mathbf{X}_k|\mathbf{Y}_{[k]})d\mathbf{X}_k \approx q_{k+1}(\mathbf{X}_{k+1}|\mathbf{Y}_{[k]}).$$

- ▶ Denoising **score matching**:

$$\hat{\mathbf{s}}_{k+1}(\cdot|\mathbf{Y}_{[k]}) = \arg \min_{\mathbf{s}:\mathbb{R} \times \mathbb{R}^d \rightarrow \mathbb{R}^d} L_{k+1}(\mathbf{s}) = \mathbb{E}_{\underline{\mathbf{X}}_{k+1} \sim \hat{q}_{k+1}(\cdot|\mathbf{Y}_{[k]})} \mathbb{E}_{\boldsymbol{\varepsilon} \sim N(0, \mathbf{I}_d)} [\|\sigma \mathbf{s}(\underline{\mathbf{X}}_{k+1} + \sigma \boldsymbol{\varepsilon}, \mathbf{Y}_{[k]}) + \boldsymbol{\varepsilon}\|_2^2]$$

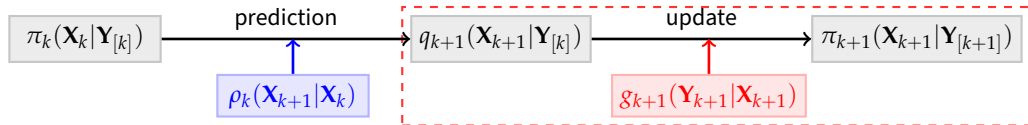
which is the score function of the Gaussian smoothed prediction distribution

$$\hat{\mathbf{s}}_{k+1}(\cdot|\mathbf{Y}_{[k]}) := \nabla \log \left(\int N(\cdot; \mathbf{X}_{k+1}, \sigma^2 \mathbf{I}_d) \hat{q}_{k+1}(\mathbf{X}_{k+1}|\mathbf{Y}_{[k]}) d\mathbf{X}_{k+1} \right) \approx \nabla \log \hat{q}_{k+1}(\cdot|\mathbf{Y}_{[k]})$$

Update via Langevin Sampling

Bayes' rule

$$\pi_{k+1}(\mathbf{X}_{k+1}|\mathbf{Y}_{[k+1]}) \propto g_{k+1}(\mathbf{Y}_{k+1}|\mathbf{X}_{k+1})q_{k+1}(\mathbf{X}_{k+1}|\mathbf{Y}_{[k]})$$



How to sample from the posterior distribution $\pi_{k+1}(\cdot|\mathbf{Y}_{[k+1]})$ given the prediction and likelihood?

$$\pi_{k+1}(\mathbf{X}_{k+1}|\mathbf{Y}_{[k+1]}) \propto g_{k+1}(\mathbf{Y}_{k+1}|\mathbf{X}_{k+1})q_{k+1}(\mathbf{X}_{k+1}|\mathbf{Y}_{[k]})$$

► Langevin dynamics

$$\begin{aligned} d\mathbf{Z}_t &= \nabla \log \pi_{k+1}(\mathbf{Z}_t|\mathbf{Y}_{[k+1]})dt + \sqrt{2}d\mathbf{B}_t, \\ &= \{\nabla \log g_{k+1}(\mathbf{Y}_{k+1}|\mathbf{Z}_t) + \nabla \log q_{k+1}(\mathbf{Z}_t|\mathbf{Y}_{[k]})\}dt + \sqrt{2}d\mathbf{B}_t, \quad \mathbf{Z}_0 \sim \hat{q}_{k+1}(\cdot|\mathbf{Y}_{[k]}). \end{aligned}$$

- Prediction score estimation
- Euler-Maruyama discretization

► Score-based Langevin Monte Carlo

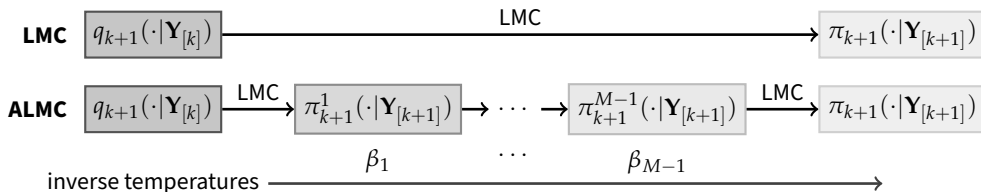
$$\hat{\mathbf{Z}}_{(k+1)h} = \hat{\mathbf{Z}}_{kh} + h\{\nabla \log g_{k+1}(\mathbf{Y}_{k+1}|\hat{\mathbf{Z}}_{kh}) + \hat{\mathbf{s}}_{k+1}(\hat{\mathbf{Z}}_{kh}, \mathbf{Y}_{[k]})\} + \sqrt{2h}\boldsymbol{\xi}_k, \quad \hat{\mathbf{Z}}_0 \sim \hat{q}_{k+1}(\cdot|\mathbf{Y}_{[k]}).$$

Annealing Strategy

Interpolations between the prediction and posterior distribution:

$$\pi_{k+1}^m(\cdot | \mathbf{Y}_{[k+1]}) \propto \pi_{k+1}(\cdot | \mathbf{Y}_{[k+1]})^{\beta_m} q_{k+1}(\cdot | \mathbf{Y}_{[k]})^{1-\beta_m}, \quad 0 \leq m \leq M$$

► Annealing schedule: $0 \equiv \beta_0 < \beta_1 < \dots < \beta_M \equiv 1$.

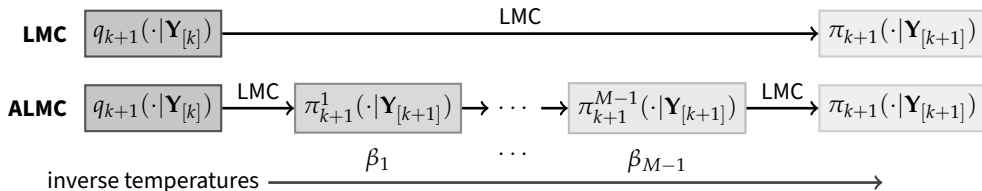


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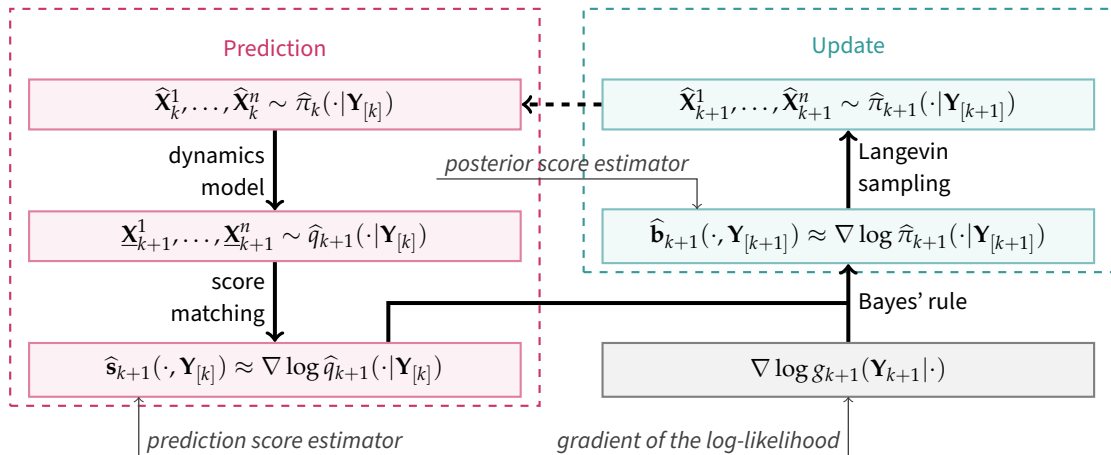
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► Annealing schedule: $0 \equiv \beta_0 < \beta_1 < \dots < \beta_M \equiv 1$.



$$\begin{aligned} \nabla \log \pi_{k+1}^m(\cdot | \mathbf{Y}_{[k+1]}) &= \beta_m \nabla \log \pi_{k+1}(\cdot | \mathbf{Y}_{[k+1]}) + (1 - \beta_m) \nabla \log q_{k+1}(\cdot | \mathbf{Y}_{[k]}) \\ &= \beta_m \nabla \log g_{k+1}(\mathbf{Y}_{k+1} | \cdot) + \nabla \log q_{k+1}(\cdot | \mathbf{Y}_{[k]}) \end{aligned}$$

Summary of Procedure



Algorithm 1: Score-based Sequential Langevin Sampling for data assimilation.

Input: The observations $(\mathbf{Y}_k)_{k \in \mathbb{N}}$, the dynamics model $(\mathcal{F}_k)_{k \in \mathbb{N}}$, the likelihood $\{g_k(\mathbf{Y}_k | \cdot)\}_{k \in \mathbb{N}}$.

Output: A particle approximation $\hat{\mathbf{X}}_{k+1}^1, \dots, \hat{\mathbf{X}}_{k+1}^n$ to the distribution $\pi_{k+1}(\cdot | \mathbf{Y}_{[k+1]})$.

```
1 # Initial posterior sampling.
2 Sample from the gauss of the initial prior distribution  $\underline{\mathbf{X}}_1^1, \dots, \underline{\mathbf{X}}_1^n \sim \text{i.i.d. } \hat{q}_1$ .
3 Estimate the score from  $\{\underline{\mathbf{X}}_1^i\}_{i=1}^n$  by score matching  $\hat{\mathbf{s}}_1$ .
4 Sample from the posterior  $\hat{\pi}_1(\cdot | \mathbf{y}_1)$  by Langevin sampling:  $\hat{\mathbf{X}}_1^1, \dots, \hat{\mathbf{X}}_1^n \leftarrow \text{ALMC}(\underline{\mathbf{X}}_1^1, \dots, \underline{\mathbf{X}}_1^n, \hat{\mathbf{s}}_1, g_1(\mathbf{Y}_1 | \cdot))$ .
5 # Recursive posterior sampling.
6 for  $k = 1, 2, \dots$  do
7     # Prediction step.
8     Run the dynamics model:  $\underline{\mathbf{X}}_{k+1}^i \leftarrow \mathcal{F}_k(\hat{\mathbf{X}}_k^i, \mathbf{V}_k^i)$  with  $\mathbf{V}_k^i \sim p_{\mathbf{V}}$  for  $1 \leq i \leq n$ .
9     Estimate the prediction score from  $\{\underline{\mathbf{X}}_{k+1}^i\}_{i=1}^n$  by score matching  $\hat{\mathbf{s}}_{k+1}(\cdot, \mathbf{Y}_{[k]})$ .
10    # Update step.
11    Sample from the posterior  $\pi_{k+1}(\cdot | \mathbf{Y}_{[k+1]})$  by Langevin sampling:
12         $\hat{\mathbf{X}}_{k+1}^1, \dots, \hat{\mathbf{X}}_{k+1}^n \leftarrow \text{ALMC}(\underline{\mathbf{X}}_{k+1}^1, \dots, \underline{\mathbf{X}}_{k+1}^n, \hat{\mathbf{s}}_{k+1}(\cdot, \mathbf{Y}_{[k]}), g_{k+1}(\mathbf{Y}_{k+1} | \cdot))$ .
13 end
14 return  $\underline{\hat{\mathbf{X}}}_{k+1}^1, \dots, \underline{\hat{\mathbf{X}}}_{k+1}^n$ 
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Assumptions

- **A1. Lipschitz score.** For each $k \in \mathbb{N}$, the posterior score is λ -Lipschitz on $\mathbb{R}^d$, that is, for each $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^d$,

$$\|\nabla \log \pi_{k+1}(\mathbf{x}_1 | \mathbf{Y}_{[k+1]}) - \nabla \log \pi_{k+1}(\mathbf{x}_2 | \mathbf{Y}_{[k+1]})\|_2 \leq \lambda \|\mathbf{x}_1 - \mathbf{x}_2\|_2.$$

- **A2. Log-Sobolev inequality.** For each $k \in \mathbb{N}$, the posterior distribution satisfies a log-Sobolev inequality with constant C_{LSI} , that is, for each function $f \in C_0^\infty(\mathbb{R}^d)$,

$$\text{Ent}(f^2) \leq 2C_{\text{LSI}} \mathbb{E}[\|\nabla f\|_2^2],$$

where the entropy is defined as $\text{Ent}(g) = \mathbb{E}[g \log g] - \mathbb{E}[g] \log \mathbb{E}[g]$, and the expectation is taken with respect to the posterior distribution $\pi_{k+1}(\cdot | \mathbf{Y}_{[k+1]})$.

► **A3. Boundedness and condition number.** For each $k \in \mathbb{N}$,

- there exists $B \geq 1$ such that $\rho_k(\mathbf{x}|\mathbf{X}_k), \|\nabla \rho_k(\mathbf{x}|\mathbf{X}_k)\|_\infty \leq B$ for $\mathbf{x} \in \mathbb{R}^d$,
- there exists $D \geq 1$ such that $q_{k+1}(\mathbf{x}|\mathbf{Y}_{[k]}) \geq D^{-1}$ for $\mathbf{x} \in \Omega_{k+1} := \text{supp}(q_{k+1}(\cdot|\mathbf{Y}_{[k]}))$,
- $q_1(\mathbf{x}), \|\nabla q_1(\mathbf{x})\|_\infty \leq B$ and $q_1(\mathbf{x}) \geq D^{-1}$ for $\mathbf{x} \in \Omega_1 := \text{supp}(q_1)$, and
- there exists $\kappa > 0$ such that

$$\frac{\sup_{\mathbf{x}} g_1(\mathbf{Y}_1|\mathbf{x})}{\int g_1(\mathbf{Y}_1|\mathbf{x})q_1(\mathbf{x})d\mathbf{x}} \frac{\sup_{\mathbf{x}} g_{k+1}(\mathbf{Y}_{k+1}|\mathbf{x})}{\int g_{k+1}(\mathbf{Y}_{k+1}|\mathbf{x})q_{k+1}(\mathbf{x}|\mathbf{Y}_{[k]})d\mathbf{x}} \leq \kappa.$$

► **A4. Error of score matching.** There exists a score matching tolerance $\Delta \in (0, 1)$ such that

$$\begin{aligned} \mathbb{E}_{\underline{\mathbf{X}}_1} [\|\nabla_{\mathbf{x}} \log \hat{q}_1(\underline{\mathbf{X}}_1) - \hat{\mathbf{s}}_1(\underline{\mathbf{X}}_1)\|_2^2] &\leq \Delta^2, \\ \mathbb{E}_{\underline{\mathbf{X}}_{k+1}} [\|\nabla_{\mathbf{x}} \log \hat{q}_{k+1}(\underline{\mathbf{X}}_{k+1}|\mathbf{Y}_{[k]}) - \hat{\mathbf{s}}_{k+1}(\underline{\mathbf{X}}_{k+1}, \mathbf{Y}_{[k]})\|_2^2] &\leq \Delta^2, \end{aligned}$$

for each $k \in \mathbb{N}$. Here the expectation $\mathbb{E}_{\underline{\mathbf{X}}_1}[\cdot]$ is taken with respect to $\underline{\mathbf{X}}_1 \sim \hat{q}_1$, and the expectation $\mathbb{E}_{\underline{\mathbf{X}}_{k+1}}[\cdot]$ is taken with respect to $\underline{\mathbf{X}}_{k+1} \sim \hat{q}_{k+1}(\cdot|\mathbf{Y}_{[k]})$.

Convergence Analysis for Posterior Sampling

Under Assumptions **A1** to **A4**. Then for each $k \in \mathbb{N}$ and each terminal time $T = Kh$,

$$\underbrace{\|\pi_{k+1}(\cdot|\mathbf{Y}_{[k+1]}) - \hat{\pi}_{k+1}(\cdot|\mathbf{Y}_{[k+1]})\|_{\text{TV}}^2}_{\text{posterior error}} \lesssim \underbrace{\exp\left(-\frac{T}{5C_{\text{LSI}}}\right)\eta_{\chi}^2}_{\text{convergence of LD}} + \underbrace{dC_{\text{LSI}}\lambda^2h}_{\text{discretization error}} + \underbrace{\sqrt{\kappa B^3 D^3}(T + C_{\text{LSI}}\eta_{\chi})\Delta}_{\text{score estimation error}} \\ + B^4 D^4 T \underbrace{\|\pi_k(\cdot|\mathbf{Y}_{[k]}) - \hat{\pi}_k(\cdot|\mathbf{Y}_{[k]})\|_{\text{TV}}^2}_{\text{prior error}},$$

where the step size h and the initial distribution $\pi_k^0(\cdot|\mathbf{Y}_{[k]})$ satisfies

$$h \lesssim \frac{1}{dC_{\text{LSI}}\lambda^2}, \quad \chi^2(\pi_{k+1}^0(\cdot|\mathbf{Y}_{[k+1]})\|\pi_{k+1}(\cdot|\mathbf{Y}_{[k+1]})) \leq \eta_{\chi}^2.$$

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- Early-stopping: Trade-off between the convergence of Langevin dynamics and score estimation error.
- Condition number: How the score matching error Δ effects the posterior error.

Convergence Analysis for Assimilation

Under Assumptions **A1** to **A4**. Suppose that the initial prior error satisfy

$$\varepsilon_{\text{init}} := \|\nabla_{\mathbf{x}} \log q_1 - \nabla_{\mathbf{x}} \log \hat{q}_1\|_{L^\infty(\mathbb{R}^d)}.$$

Then for each time step $k \in \mathbb{N}$ and error tolerance $\varepsilon \in (0, 1)$,

$$\|\pi_{k+1}(\cdot | \mathbf{Y}_{[k+1]}) - \hat{\pi}_{k+1}(\cdot | \mathbf{Y}_{[k+1]})\|_{\text{TV}} \leq \tilde{\mathcal{O}}(\varepsilon_{\text{init}} + \varepsilon),$$

where the $\tilde{\mathcal{O}}$ notation omits logarithmic factors of $\varepsilon_{\text{init}}$ and ε , and the constant behind the $\tilde{\mathcal{O}}$ notation is independent of $\varepsilon_{\text{init}}$ and ε . Moreover, the step size h , the number of the Langevin iterations K and the score matching error Δ satisfy:

$$h = \Theta\left(\frac{\varepsilon^2}{dC_{\text{LSI}}\lambda^2}\right), \quad K = \Theta\left(\frac{d\lambda^2C_{\text{LSI}}^2}{\varepsilon^2} \log\left(\frac{\eta_\chi^2}{\varepsilon^2}\right)\right),$$
$$\Delta = \Theta\left(\frac{\varepsilon^2}{\sqrt{\kappa}B^3D^3C_{\text{LSI}}} \frac{1}{\log(\varepsilon^{-2}\eta_\chi^2) + \eta_\chi}\right).$$

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Double-Well Potential

Dynamics model:

$$dX_t = -\nabla U(X_t)dt + \beta dB_t$$

- Double-well potential: $U(x) = x^4 - 2x^2$.

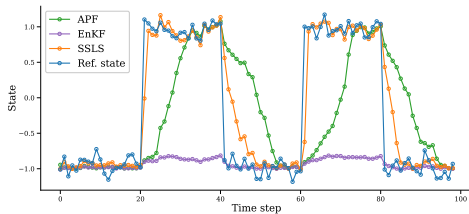
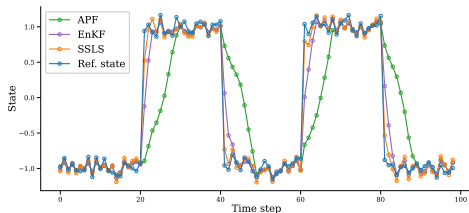
Measurement model:

- Linear measurement model:

$$Y_k = X_k + \sigma W_k$$

- Nonlinear measurement model:

$$Y_k = \exp(X_k - \gamma_k) + \sigma W_k$$



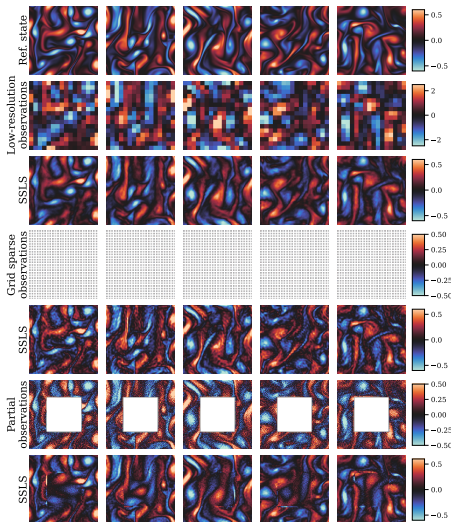
Kolmogorov flow

Dynamics model:

$$\begin{cases} \partial_t \mathbf{u} = -(\mathbf{u} \cdot \nabla) \mathbf{u} + \frac{1}{\text{Re}} \nabla^2 \mathbf{u} - \frac{1}{\rho} \nabla p + \mathbf{F}, \\ 0 = \nabla \cdot \mathbf{u}. \end{cases}$$

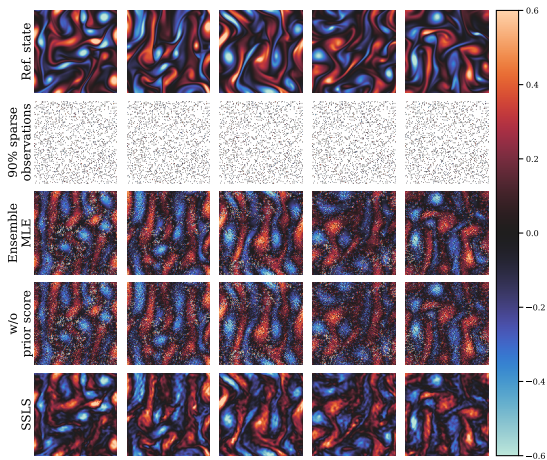
Measurement model:

- ▶ Low-resolution observation
- ▶ Sparse observation
- ▶ Partial observation

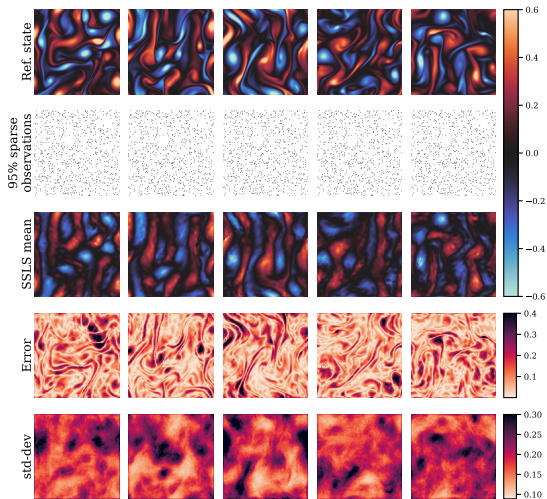
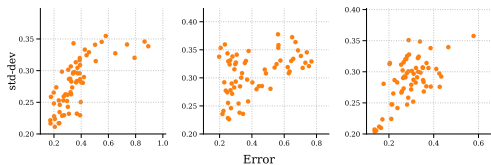


Ablation study: influence of the prediction score

- ▶ Score-based Langevin sampling
- ▶ Langevin sampling without prediction score
- ▶ Ensemble maximum likelihood estimation



Uncertainty quantification



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Concluding Remarks

- ▶ Improvement of the computational efficiency.
- ▶ Provable non-log-concave posterior sampling.
- ▶ Refined error bounds for long-time assimilation.
- ▶ Applications in the real data.

Reference: Zhao Ding, Chenguang Duan, Yuling Jiao, Jerry Zhijian Yang, Cheng Yuan, and Pingwen Zhang. Nonlinear Assimilation via Score-based Sequential Langevin Sampling. `arXiv:2411.13443`.

Thanks for your attention!

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